Lorenzo Masoero

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Senior Applied Scientist (AS) | Amazon, Weblab

September 2021 - Present (*09/21–09/22: AS I; 09/22–05/24: AS II; 05/24–current: AS III*)
I am a Senior Applied Scientist at Weblab, Amazon's largest A/B testing platform, where I work on advancing online experimentation at scale. My work spans across open-ended research, academic paper writing, and the hands-on implementation of large-scale algorithms in production, directly impacting hundreds of millions of customers daily. My main focus in Weblab has been:

> **Online Experimentation with Interference.** Pioneered "multiple randomization designs'' at Amazon (joint work with Prof. Guido Imbens and Prof. Thomas Richardson, amongst others)—experiments that jointly randomize across multiple marketplace units to detect spillover effects invisible to classical A/B tests. This line of work resulted in Multiple Randomization Designs (JRSS-B, 2026), Experimental Design in Marketplaces (Statistical Science, 2023), Multi-Armed Bandits with Network Interference (NeurIPS 2024), and Efficient Switchback Experiments via MRDs (CODE@MIT 2023), amongst others.
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> **Advancing Experimentation Methodology.** Beyond interference, I extend standard A/B testing methodology across several fronts. I designed and deployed a distributed PySpark engine for covariate-adjusted causal estimation at scale (KDD CDPD, 2023) serving thousands of daily experiments. With Suhas Vijaykumar and Prof. Imbens, I developed Demonstration Experiments (pre-print, 2025)—adaptive trials designed to prove that at least one intervention works, formalized as a bandit problem with valid inference under adaptive allocation. With Paul Missault, I work on robust switchback designs (CODE@MIT 2025) for time-randomized experiments. I also develop Bayesian nonparametric methods for traffic prediction in online experiments (Annals of Applied Statistics, to appear).
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> **Agentic Workflows for Experimentation Pipelines.** Leading integration of LLM-powered agents into experimentation infrastructure. In one direction, I built a multi-agent system (TriAgent, IAAI 2027) that identifies documentation gaps from support interactions and self-improves over time. In a second direction, I am developing autonomous LLM experimenters (Agentic Experimentation, AMLC 2026 Workshop) that interpret metrics, diagnose outcomes, and propose next steps—reducing the human bottleneck in large-scale A/B testing pipelines.

Previously, I completed my PhD at MIT in EECS in August 2021 under the supervision of Professor Tamara Broderick. During my PhD, I worked on Bayesian inference, with an emphasis on the development of efficient and theoretically grounded approaches for optimal design of experiments in genomic studies. In particular, my PhD thesis consisted of a sequence of papers developing a methodology to devise optimal sampling strategies to maximize learnings stemming from rare-variants discoveries under fixed experimental budgets using a Bayesian nonparametric approach ("More for Less: Predicting and maximizing genetic variant discovery via Bayesian nonparametrics", in Biometrika, "Scaled process priors: Improved predictions and uncertainties for new-feature counts via random scaling in Bayesian nonparametrics", in the Journal of the American Statistical Association, "Bayesian nonparametric strategies for power maximization in rare variants association studies", spotilight at Learning Meaningful Representations of Life, Neural Information Processing Systems, 2021).

Employment

2024- Applied Scientist III, Weblab, Amazon.
2022-2023 Applied Scientist II, Weblab, Amazon.
2021 Applied Scientist, Weblab, Amazon.
2020 Applied Science Intern, CoreAI, Amazon.

Education

2016-2021 PhD, Electrical Engineering and Computer Science, Massachusetts Institute of Technology
2015 - 2016 MA in Statistics and Applied Mathematics, with distinction, Collegio Carlo Alberto
2015 - 2016 MA in Quantitative Finance, 110/110 magna cum laude, Università degli Studi di Torino
2012 - 2014 BA in Economics, 110/110 cum laude, Università degli Studi di Torino

Scholarships and Awards

2020 SBSS Best Student Paper Award (ASA)
2020 Bayes Comp Travel Award
2018 BNP@NeurIPS Award
2017 Andrew (1956) and Erma Viterbi Fellowship
2016 Best Graduate Student of the Year (ATLEC)
2015 - 2016 Graduate Allievi Honors Program Scholarship, Collegio Carlo Alberto, Moncalieri
2012 - 2014 Undergraduate Allievi Honors Program Scholarship, Collegio Carlo Alberto, Moncalieri

Research

- "**Can AI Agents Simulate A/B Test Outcomes? A Validation Framework for Agentic Experimentation**"; COLM 2026 Workshop on Agent Behavior; CODE@MIT 2026. (Hut, **M**.)
- "**Mind the Gap: A Deployed Multi-Agent System for Documentation Maintenance**"; AAAI Conference on Innovative Applications of Artificial Intelligence (IAAI) 2027. (**M**.)
- "**From Offline Evidence to Online Action: A Decision Framework for Imperfect Offline Evaluation**"; ICML 2026 Workshop on Decision-Making from Offline Datasets to Online Adaptation. (Wang, **M**.)
- "**Carryover detection in switchback experimentation**"; CauScien Workshop, NeurIPS 2025. (Missault, **M**.)
- "**Optimizing duration of online experiments via Bayesian early termination**"; Spotlight talk at Code@MIT 2025. (**M**., Gualavisi)
- "**Rethinking Power in Online Experimentation: Connecting Frequentist and Bayesian Approaches to Inform Experiment Design**"; Spotlight talk at Code@MIT 2025. (Gualavisi, Kessler, **M**.)
- "**Robust and efficient multiple-unit switchback experimentation**" Code@MIT 2025. (Missault, **M**., Delbé, Richardson, Imbens)

- "**Online activity prediction via generalized Indian buffet process models**" Annals of Applied Statistics, to appear. (Beraha, **M.**, Favaro, Richardson)
- "**Measuring direct and Indirect Impacts in a Multi-Sided Marketplace: Evidence from a Clustered Multiple Randomization Experiment**" CODE@MIT2024, KDD 2024, Workshop on Two-Sided Marketplace Optimization; (Hut, Imbens, **M.**, Vijaykumar)
- "**Multi-Armed Bandits with Network Interference**"; NeurIPS 2024. (Agarwal, Agarwal, **M.**, Whitehouse)
- "**Double trouble: Predicting new variant counts across two heterogeneous populations**"; arXiv preprint, 2403.02154. (Shen, **M.**, Schraiber, Broderick)
- "**Multiple Randomization Designs: Estimation and Inference with Interference.**" Journal of the Royal Statistical Society Series B: Statistical Methodology, 2026. (**M.**, Vijaykumar, Richardson, McQueen, Rosen, Burdick, Bajari, Imbens). Manuscript available at arXiv
- "**Leveraging covariate adjustments at scale in online A/B testing**"; 2023 KDD Workshop on Causal Discovery, Prediction and Decision, PMLR 218:25-48, 2023. (**M.**, McQueen, Hains)
- "**Experimental Design in Marketplaces**"; Statistical Science, 38(3): 458-476 (August 2023). (Bajari, Burdick, Imbens, **M.**, McQueen, Richardson, Rosen). Manuscript available at Amazon Science.
- "**Independent Finite Approximations for Bayesian Nonparametric Inference**"; Bayesian Analysis, Advance Publication 1-38 (2023). (Nguyen, Huggins, **M.**, Mackey, Broderick)
- "**Double trouble: Predicting new variant counts across two heterogeneous populations**"; Learning Meaningful Representations of Life, Neural Information Processing Systems Workshop, 2022. (Shen, **M.**, Schraiber, Broderick). [Preprint available on OpenReview]
- "**Cross-Study Replicability in Cluster Analysis**"; Statistical Science, 38(2): 303-316 (May 2023). (**M.**, Thomas, Parmigiani, Tyekucheva, Trippa)
- "**Multiple Randomization Designs**"; Amazon Science, 2021. (Bajari, Burdick, Imbens, **M.**, McQueen, Richardson, Rosen)
- "**Bayesian nonparametric strategies for power maximization in rare variants association studies**"; **Spotlight** at Learning Meaningful Representations of Life, Neural Information Processing Systems, 2021 [poster]. Manuscript available on arXiv [https://arxiv.org/pdf/2112.02032.pdf] (**M.**, Schraiber, Broderick)
- "**The Chicago Police Department dataset**"; Neural Information Processing Systems Track on Datasets and Benchmarks, 2021, Dataset repository, current release (v0.1) and documentation (Horel, **M.**, Agrawal, Roithmayr, Campbell)
- "**Scaled process priors: Improved predictions and uncertainties for new-feature counts via random scaling in Bayesian nonparametrics**"; Accepted for publication in the Journal of the American Statistical Association. Manuscript available on arXiv [poster; https://arxiv.org/pdf/2106.15480.pdf]. Featured in ISBA 2021; (Camerlenghi, Favaro, **M.**and T. Broderick)
- "**More for Less: Predicting and maximizing genetic variant discovery via Bayesian nonparametrics**"; to appear in Biometrika. **Best Student Paper Award**, awarded by the American Statistical Association, Section in Bayesian Statistical Science. Featured in SMEEB

2021, ASHG 2020, AABI 2019 [poster; presentation (YouTube)]; Manuscript available on arXiv [https://arxiv.org/pdf/1912.05516.pdf] (**M.**, Camerlenghi, Favaro, Broderick)

- "**Posterior representations of hierarchical completely random measures in trait allocation models**", **Spotlight**, *BNP@NeurIPS2018* [poster] (**M.**, Camerlenghi, Favaro and Broderick)
- "**Sensitivity of Bayesian inference to data perturbations**", *AABI 2018* [poster] (**M.**, Stephenson, Broderick)
- "**Generic finite approximations for practical Bayesian nonparametrics**", **Spotlight**, *NIPS 2017 Workshop on Advances in Approximate Bayesian Inference* [poster]. (Huggins, **M.**, Mackey, Broderick)

Theses

- "**Improved prediction and optimal sequencing strategies for genomic variant discovery via Bayesian nonparametrics**" — PhD thesis. Supervisor: Professor Tamara Broderick
- "**An asymptotic analysis of Gibbs-type priors**" — Master's thesis. Supervisors: Professors Pierpaolo de Blasi and Igor Prünster
- **"Econometrics of the Big Data"** — Undergraduate thesis. Supervisor: Professor Alessandro Sembenelli

Skills

- Proficient in Python (numpy, scipy, pandas, matplotlib, scikit-learn), LaTeX
- Past experience in C++, Matlab, R, RStudio

Talks, Poster sessions and Conference Presentations

2025

- Conference on Online Digital Experimentation, MIT. Three selected papers: two selected talks, one poster.
- NeurIPS; Poster presentation at https://sites.google.com/view/causcien.

2024

- Conference on Online Digital Experimentation, MIT. Three selected papers: two selected talks, one poster.
- KDD 2024 Workshop on Two Sided Marketplace Optimization, selected talk.
- Conference on Online Digital Experimentation, MIT, two selected talks, one poster session.

2023

- KDD Workshop on Causal Discovery, Prediction and Decision, Long Beach (CA).
- Conference on Online Digital Experimentation, MIT, selected talk.

2022

- Conference on Online Digital Experimentation, MIT, selected talk.

2021

- Collegio Carlo Alberto Statistics Seminar Series, "Improved prediction and optimal sequencing strategies for genomic variant discovery via Bayesian nonparametrics"
- CMS Statistics 2021, "Scaled process priors for Bayesian nonparametric estimation of the unseen genetic variation" [Invited session]
- ISBA: 2021 World Meeting of the International Society for Bayesian Analysis, "Scaled process priors for Bayesian nonparametric estimation of the unseen genetic variation" [Contributed session]
- SMEEB: Stochastic Models and Experiments in Ecology and Biology, "More for less: predicting and maximizing genomic diversity via Bayesian nonparametrics" [Contributed session]

2020

- American Society of Human Genetics meeting, "More for less: predicting and maximizing genomic diversity via Bayesian nonparametrics" [Poster session]
- Learning under complex structure, MIFODS workshop, *Cambridge (MA)*, "More for less: predicting and maximizing genomic diversity via Bayesian nonparametrics" [Poster session]

- Learning under complex structure, MIFODS workshop, *Cambridge (MA)*, "More for less: predicting and maximizing genomic diversity via Bayesian nonparametrics" [Poster session]
- Bayes Comp 2020, *Gainesville (FL)*, "More for less: predicting and maximizing genomic diversity via Bayesian nonparametrics" [Poster session]

2019

- Advances in Bayesian Nonparametric Methods and Its Applications, *Denver (CO), JSM 2019*, "Genomic variety prediction via Bayesian nonparametrics" [Topic-contributed session]
- Advances in Approximate Bayesian Inference, *Vancouver, Canada*, "More for less: Predicting and maximizing genetic variant discovery via Bayesian nonparametrics"
- Statistics and Data Science Conference 2019, *Cambridge (MA)*. "Genomic variety prediction via Bayesian nonparametrics"
- MLxMIT, *Cambridge (MA)*, "Genomic variety prediction via Bayesian nonparametrics"
- LIDS *&* Stats seminar, *Cambridge (MA)*, "Genomic variety prediction via Bayesian nonparametrics"
- CSAIL-MSR Trustworthy and Robust AI (TRAC) Workshop, *Cambridge (MA)*, "Getting the most bang for your buck: Predicting and maximizing the number of new genetic variants in a future experiment"

2018

- BNP@NeurIPS 2018, Montreal (Canada) "Posterior representations of hierarchical completely random measures in trait allocation models" [**Spotlight**]

Professional Service

2026 Reviewer for ProbML, AISTATS, NeurIPS, KDD
2025 Reviewer for JASA, KDD, AISTATS, NeurIPS
2024 Reviewer for KDD, AISTATS, AABI
2023 Reviewer for JMLR, AISTATS, DGM4H@NeurIPS2023
2022 Reviewer for Statistical Science, JMLR, AISTATS
2021 Reviewer for Statistical Science, AISTATS, ICBINB
2020 Reviewer for AAAI 2020, AISTATS 2020
2019 Reviewer for AISTATS 2019, NeurIPS 2019, AABI 2019
2018 Reviewer for BNP@NeurIPS2018